

Predicting Soccer Player's Market Value

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Abstract—Understanding soccer players' market values is crucial in the modern football ecosystem. Accurate predictions can assist clubs in making data-driven decisions during transfer negotiations, helping them determine fair pricing for acquisitions and sales. Agents knowing how various factors impact player valuation can inform contract negotiators and career strategies. Soccer fans and analysts can also benefit from a clearer understanding of player market dynamics, enriching discussions around player performance and team composition.

This project investigates the prediction of soccer players' market values using a dataset containing detailed information on 17,000 players and 51 features. By framing the problem as a regression task, the project aims to determine how attributes such as age, skill ratings, wages, and positions influence market valuation. Multiple machine learning models were employed, including linear regression, random forest, Gradient boosting, and neural networks, with extensive parameter tuning and cross-validation to ensure robust predictions. Data preprocessing steps included handling missing values, outliers, and feature scaling. The random forest model achieved the best performance with a Mean Absolute Error (MAE) of 0.05 and an R^2 score of 1.00, significantly outperforming the neural network model. Insights gained from feature importance analysis further clarified the factors most strongly associated with player valuation. While challenges such as handling outliers and incorporating inflation adjustments were addressed, the findings demonstrate that tree-based models excel in this regression task, providing a reliable framework for market value prediction in professional soccer.

Index Terms—Linear Regression, Random Forest, Neural Networks, Prediction, Supervised Learning, Ensembles, Sci-kit-learn, TensorFlow, Soccer Player Market Value, Soccer

I. INTRODUCTION

In the world of professional soccer, a player's market value serves as a critical indicator of their skill, potential, and reputation. Accurate estimation of market values is essential for clubs to make informed decisions about transfers, contract negotiations, and overall team strategy. However, the valuation process is inherently complex, influenced by a wide array of factors ranging from a player's age and physical attributes to their performance metrics and international reputation. This project addresses the challenge of predicting soccer players' market values by leveraging machine learning techniques to uncover patterns and relationships within a rich dataset.

The task of market value prediction is framed as a regression problem, where the goal is to estimate a continuous variable—the player's market value in euros. Unlike classification tasks, which categorize data into discrete labels, regression

requires modeling complex relationships across multiple numerical and categorical features. Prior studies in this domain have explored various modeling approaches, highlighting the potential of data-driven methods to augment traditional expert-driven valuations. This project employs several advanced machine learning algorithms—including linear regression, random forest, Gradient boosting, and neural networks—to develop a robust predictive framework. Each model is evaluated for its performance, and the best-performing model is selected based on metrics such as Mean Absolute Error (MAE) and R^2 score. Additionally, I create a custom deep learning model and examine the results to explore whether this approach provides further predictive power.

The project incorporates data preprocessing techniques, such as feature scaling and handling missing values, to ensure model reliability. The integration of feature importance analysis further elucidates the attributes most strongly associated with market value, offering valuable insights into the key drivers of player valuation. Notably, features such as age, overall rating, potential, finishing, and dribbling emerged as critical factors, reflecting both a player's immediate capabilities and long-term potential to contribute effectively on the field.

This study not only advances the application of machine learning in sports analytics but also provides actionable insights for soccer clubs and analysts, enabling them to make data-driven decisions in player acquisitions, contract negotiations, and talent scouting based on identified key drivers such as age, overall rating, and potential. By comparing the performance of multiple models and fine-tuning their parameters, the project establishes a comprehensive methodology for market value prediction, laying the groundwork for future enhancements and applications.

II. BACKGROUND

Soccer players' market value is a multifaceted measure, reflecting their on-field performance, physical attributes, and perceived potential. Predicting market value is crucial for various stakeholders, including clubs, agents, and analysts. Clubs rely on accurate valuations to strategize player acquisitions and transfers effectively. Agents use these valuations to negotiate contracts and highlight their clients' worth. Accurate market value predictions also empower analysts to track trends and identify undervalued players, providing a competitive edge in a highly dynamic market.

Among the key features influencing market value, overall performance metrics, such as overall rating and potential, play a pivotal role. A player's overall rating captures their current capabilities, summarizing technical, physical, and mental attributes. Potential, on the other hand, provides a forward-looking perspective, quantifying a player's growth capacity and long-term utility for a club. For instance, young players with high potential often command higher market values due to their perceived future impact.

Technical skills, including finishing, volleys, dribbling, and positioning, directly correlate with a player's effectiveness on the field. For attacking players, attributes like finishing and volleys determine their goal-scoring proficiency, a critical factor in market valuation. Similarly, dribbling reflects a player's ability to maneuver in tight spaces, while positioning indicates their tactical awareness and adaptability to game scenarios. These features collectively highlight a player's offensive contributions, making them invaluable for clubs seeking goal-oriented talent.

Defensive skills, such as sliding tackle, are equally important, especially for defenders whose primary role is to prevent the opposing team from scoring. This attribute reflects a player's ability to intercept plays and maintain defensive stability, both of which significantly enhance their market value.

Physical and derived metrics, including age to potential ratio and performance index, offer unique insights into a player's career trajectory. The age to potential ratio evaluates how close a player is to realizing their full potential, providing clubs with a sense of urgency regarding development opportunities. The performance index, calculated as a composite of key attributes like overall rating, potential, and composure, serves as a holistic measure of a player's contributions, offering a straightforward metric for comparison.

We made sure to focus on these features so we can ensure that the prediction models align with real-world decision-making processes in professional soccer. Berrar, D., Lopes, P. and Dubitzky, W (2019) stated, "key factor in soccer match outcome prediction lies in the successful incorporation of domain knowledge into the machine learning modeling process." [1] Although this is relevant to soccer match outcome prediction, it should also correlate with the player's market value prediction. Knowing the data and the ability to interpret and prioritize these features not only improves the accuracy of market value predictions but also provides actionable insights for stakeholders across the soccer ecosystem. We also made sure to run tests that do not involve selecting attributes via domain knowledge. We wanted to see any hidden correlations, but that resulted in similar importance.

III. EXPERIMENTAL METHODOLOGY

The experiment methodology details the systematic steps taken to prepare the dataset, select and fine-tune models, and evaluate their performance. This section also highlights the tools and libraries leveraged in this experiment. They include popular machine learning libraries such as Scikit-learn [2]

for model training, TensorFlow [3] for more complex neural network-based approaches, and Matplotlib [4] for visualizing data patterns. Various machine learning models were applied, including traditional methods like linear regression, advanced ensemble models such as Random Forest, and deep learning approaches like neural networks.

A. Dataset Description

The dataset [5] used for this experiment was sourced from Kaggle, which collected player data from SoFIFA.com, a trusted platform for detailed soccer statistics. The data covers professional soccer players worldwide, collected between October 18, 2023 and October 31, 2023, with 17,000 records and 51 features. These features encapsulate a broad array of player characteristics such as age, skill ratings, physical attributes, and financial data, like market value and wage. The data's comprehensive nature allows for robust analysis and model building, making it suitable for predicting the value of players based on multiple factors.

B. Data Preprocessing

Several preprocessing steps were performed to ensure the dataset's suitability for machine learning tasks:

1) *Handling Missing Values:* Records with missing values in critical features, such as `value_euro` (market value), were removed, as the absence of a target variable for training would make it impossible to generate meaningful predictions. Managing missing features is crucial because improperly handling them can skew results, introduce noise, and compromise the reliability of the models, especially in tasks requiring consistent and complete data for accurate learning. We also handled features that were mainly present in the dataset. This included the player's national team. Most of it was missing, so instead of deleting records with that feature value missing, we dropped the column entirely by not taking it into consideration. We also only focused on numerical features for our models, and the national team feature was a string feature. We checked if it had any significance by looking at just records with the national teams, which were few, but there weren't strong correlations between market values anyway.

2) *Scaling and Normalization:* Features like `height_cm`, `weight_kgs`, and numerical skill ratings exhibited different scales and units. For example, height is measured in centimeters, while wage is expressed in euros, which could cause disparities in how much each feature influences the model. To address this, the features were standardized using z-score normalization to ensure all variables contributed equally to the model training process. This method standardizes the data by transforming it to have a mean of zero and a standard deviation of one. Normalization ensures that all features contribute equally during model training, particularly when using algorithms like linear regression that are sensitive to feature scaling. This prevents features with larger scales or units from disproportionately influencing the model, thereby improving the fairness and reliability of predictions.

3) *Outlier Handling*: Outliers, particularly in high-value features like wage euro and value euro, were identified but retained due to the robustness of tree-based models like Random Forest. These models are less sensitive to extreme values, unlike algorithms like linear regression, which may require more stringent outlier removal. For linear regression, we still decided not to touch outliers since they mainly existed in only a few features, mainly for wage euro. Later in the paper, we will talk about the specific results, but we did a multitude of tests.

4) *Target Transformation*: The value_euro feature, representing the market value of players, showed significant right skewness in its distribution. In such cases, a log transformation was applied to the target variable. The reason for this transformation is twofold: first, to address the skewness in the distribution of market values, and second, to reduce the impact of extremely high values that could unduly influence the model. By taking the logarithm of value_euro, the resulting distribution becomes more symmetric, making the model more stable and capable of making accurate predictions. This transformation also helps with model convergence during training, improving efficiency and accuracy in the predictions of market values. The log-transformed target is referred to as value_euro_log.

C. Feature Engineering

Feature engineering plays a crucial role in improving the predictive power of machine learning models. In this study, several new features were created to help the model better understand the underlying relationships between player attributes and their market value:

- 1) value_euro_log: This feature represents the log-transformed version of the original market value. As mentioned earlier, this transformation was performed to reduce the skewness and help the model better generalize when predicting values for players with extremely high market values. It allows the model to handle a broader range of data more effectively, improving overall accuracy.
- 2) performance_index: A new composite feature was introduced, the performance_index, which is calculated by averaging three key attributes: overall_rating, potential, and composure. This index provides a quick overview of a player's general performance, accounting for their rating, potential for growth, and mental attributes like composure. This feature helps the model by providing a single metric that can be correlated with a player's market value rather than relying on each of these features separately.
- 3) age_to_potential_ratio: This derived metric assesses how close a player is to realizing their potential by calculating the ratio of age to potential. Players with a low ratio (young players close to their potential) are often viewed as more valuable investments. On the other hand, players with higher ages and lower potential may have lower market values. This feature provides a comparative

understanding of a player's development stage, which is crucial for accurate market value predictions.

By creating these features, we aim to provide the model with more meaningful inputs that could potentially uncover hidden patterns and improve prediction accuracy. It was decided not to implement inflation calculations against market value in our dataset. Instead, it would be better to directly re-scrape from the so-fifa.com website. The website automatically accounts for inflation. This data is associated with October 2023. However, if we have data that is associated with a different time period that we want to test, then inflation metrics need to be done using the inflation formula. We get the inflation index from another data set and apply it to the market value to get an adjusted market value. Yigit, Ahmet Talha, Samak, Baris, and Kaya, Tolga(2020) provided a very nice implementation in their papers which we could also utilize. [6]

$$\text{Adjusted Market Value} = \text{Market Value} \times \frac{\text{Inflation Index (Base Year)}}{\text{Inflation Index (Current Year)}} \quad (1)$$

IV. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is an essential step in understanding the structure and distribution of data. EDA helps in identifying patterns, relationships, and anomalies within the dataset, which can inform the choice of modeling techniques. We used two primary analysis techniques: univariate and bivariate.

1) *Univariate Analysis*: This analysis focuses on examining the distribution of individual variables. We took all the numerical features and graphed their frequencies in the data. We could visualize the distributions of key features like value_euro and overall_rating. Histograms and boxplots showed skewness for features like value_euro(market value). We also saw which ranges of features appeared the most in our dataset. For instance, the value_euro feature displayed a right-skewed distribution, which led us to apply the log transformation discussed earlier. Additionally, the overall rating feature was relatively uniform, indicating a less pronounced skew and suggesting that it may be a stable predictor.

2) *Bivariate Analysis*: Bivariate analysis examines the relationships between two features. We used scatter plots to investigate how the value_euro feature correlates with other important features. We found intriguing correlations between multiple features. There were clean graphs showing direct correlations between potential and market value. It is a sharp exponential increase in market value as potential is above 80. Overall rating also had similar results. We also created a heatmap to compare our important features. In that case, we compared a multitude of features with each other. The more red they are, the more closely related they are. Of course, the lines i and jth entries with the same values have the highest rating because they are being mapped with themselves. However, we can see the other features that are more important. The heatmap could also help identify potential issues like multicollinearity, where two or more features are highly correlated, which could impact model performance.

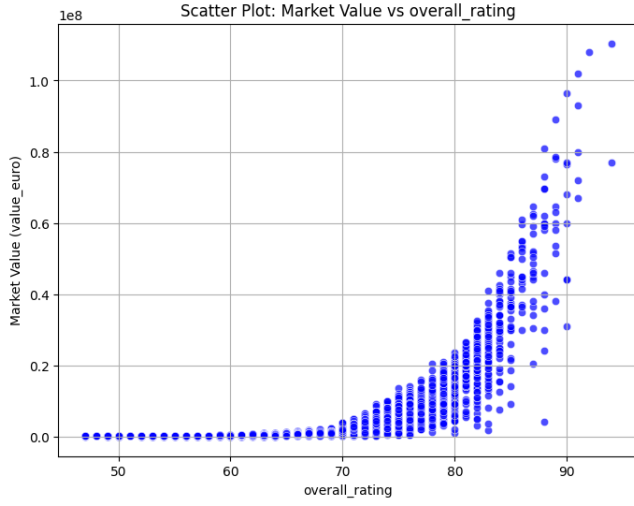


Fig. 1. Scatter Plot

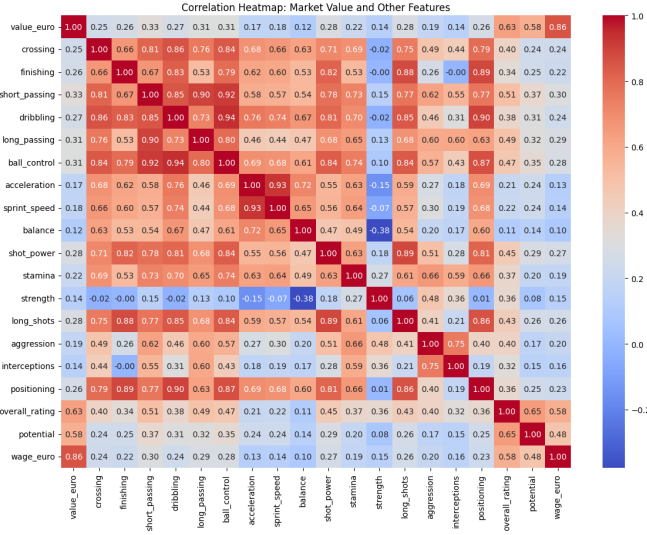


Fig. 2. Correlation Matrix Heat Map

- Scatterplots were generated to examine relationships between value_euro and other features, such as potential and wage_euro, demonstrating strong positive correlations.
- A correlation heatmap was created to identify multicollinearity and key feature relationships, confirming the importance of attributes like overall rating and potential in predicting market value.

V. SETUP METHODOLOGY

The evaluation strategy involved mean absolute error (MAE), which is a measure of average error magnitude; root mean squared error (RMSE), which is sensitive to larger errors; and R^2 score, which quantifies the variance explained by the model. [7]

$$MAE = \frac{\sum_{i=1}^n |y_i - x_i|}{n} \quad (2)$$

y_i : true value
 x_i : predicted value
 n : total number of data points

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{n}} \quad (3)$$

$\hat{y}_i$: predicted value
 y_i : true value
 n : total number of data points

$$R = 1 - \frac{RSS}{TSS} \quad (4)$$

R : coefficient of determination
 RSS : sum of squares of residuals
 TSS : total sum of squares

The dataset was split into a training set (66%) and a test set (34%). Additionally, a validation set was created during cross-validation for hyperparameter tuning. K-fold cross-validation (with k=5) was employed to ensure model robustness and minimize overfitting. This approach divides the training data into five folds, where four folds are used for training, and the fifth is reserved for validation. This process is repeated five times, with each fold used exactly once as the validation set. To establish a baseline, a simple linear regression model was implemented without regularization or feature engineering. This allowed for a comparison of complex models against a fundamental approach. Additionally, benchmarks were set using industry standards and prior works on similar datasets. The experiment utilized several libraries in python. Scikit-learn for classical machine learning models, parameter tuning, and evaluation metrics. TensorFlow/Keras is used to design and train neural networks. Matplotlib and Seaborn [8] for data visualization and exploratory analysis.

VI. MODELS AND PARAMETER TUNING

We started off with linear regression from scikit-learn. It served as a straightforward benchmark model. Variants like Lasso and Ridge regression were also explored. Miao He, Ricardo Cachucho, and Arno Knobbe(2015) explored how Lasso Regression can be a key metric to apply against predictive numerical features, especially in player market value. [9] Lasso (Least absolute shrinkage and selection operator) and Ridge regression are both types of linear regression models. They apply regularization to prevent overfitting since our dataset was quite large with a moderate amount of features and

seemed to face multicollinearity with features that we created (age_to_potential_ratio).

Ridge:

$$L(\beta) = \left\{ \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^p \beta_j^2 \right\} \quad (5)$$

Lasso:

$$L(\beta) = \left\{ \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^p |\beta_j| \right\} \quad (6)$$

$L(\beta)$: Loss/objective function to be minimized
 y_i : Actual observed values (target variable)
 $\hat{y}_i$: Predicted values from the model
 $\sum_{i=1}^n$: Sum over all n training examples
 α : Regularization parameter controlling regularization strength
 β_j : Model coefficients/parameters
 $\sum_{j=1}^p$: Sum over all p features

A custom linear regression was made as well to test the performance of including and not including unnecessary features. As stated by Weiss [10], unnecessary features can be detrimental to model performance in models like linear regression. Copying most of the original scikit-learn capabilities, this linear regression used the closed-form solution. The closed-form solution computes the weights or the coefficients of the linear model directly.

$$w = (X^T X)^{-1} X^T y \quad (7)$$

w : Vector of model weights
 X : Matrix of selected features, including a bias column of 1s for intercept
 y : Target variable (e.g., market value)

Then Random Forest, a tree-based ensemble method was utilized. It builds a collection of decision trees and merges their results to improve the accuracy and generalization of predictions. Our task was a regression task, so a random forest would be a good candidate model to use. It was also utilized for its ability to model non-linear relationships. Random Forest was tuned to see whether changing hyperparameters can improve the model's performance. The goal was finding the best combination of hyperparameters which minimizes error.

TABLE I
PARAMETER GRID FOR RANDOM FOREST

Parameter	Values
Trees	[100, 200]
max depth	[None, 10, 20]
minimum samples split	[2, 5]
min samples leaf	[1, 2]
max features	['sqrt', 'log2', None]

GridSearchCV was employed to identify optimal values, with performance evaluated on validation sets.

Next XGBoost was chosen for its efficiency and accuracy in regression tasks. We did not fine-tune Gradient boosting. We took into account potential benefits of fine-tuning and what could go wrong. Gradient boosting itself is an ensemble learning technique. It builds a predictive model by combining multiple weak learners. Scikit-learn uses decision trees. The weak learners correct the errors of the previous weak learners that came before them. The boosting is supposed to improve the performance of a weak learner by combining it with other learners, but the key difference between other boosting methods is that gradient descent optimization techniques were used to minimize the loss function in a sequential manner.

TABLE II
PARAMETER GRID FOR XGBOOST

Parameter	Values
Trees	100
learning rate	0.1
max depth	3
random state	default(42)

A Multi-Layer Perceptron (MLP) regressor was implemented to evaluate the potential of deep learning. This is a type of neural network which consists of multiple layers of nodes. Composed of an input layer, one or more hidden layers, and an output layer. These networks are supposed to be capable of learning complex patterns through the training process, where the model adjusts the weights of connections to minimize the error in predictions. They use algorithms like backpropagation and gradient descent. The architecture of the MLP consisted of several key components. They were selected to provide optimal model performance. The input layer was configured to match the number of features in the dataset. The hidden layers and associated neurons in each varied in the configurations as stated in table 3. We used the rectified linear unit for hidden layers to introduce non-linearity and linear for the output layer. We also used the adam optimizer for training and configured the learning rates for smooth learning. This combination of techniques provided a robust foundation for training the MLP and improving its generalization capabilities, but the results varied.

TABLE III
PARAMETER GRID FOR NEURAL NETWORK FINE TUNING

Parameter	Values
hidden layer sizes	[(100.), (50, 50), (100, 50, 25)]
activation	['relu', 'tanh']
solver	['adam']
learning rate init	[0.0005, 0.001]
alpha	[0.0001, 0.001, 0.01]
max iterations	[500]

Deep learning was incorporated as well. Where a sequential model was utilized using TensorFlow Keras API. Reference to V. Steve Arrul, P. Subramanian and R. Mafas (2022) was used as motivation for the initialization of the layers. [11] It consisted of three layers. The input layer corresponded to the number of features in the dataset. The first hidden layer

consists of 128 neurons with ReLU activation. This introduced nonlinearity to the model. A dropout layer with a rate of 0.2 was introduced after the first hidden layer to randomly deactivate 20% of the neurons during training. This forces the networks to learn more robust features and prevents it from overly relying on specific neurons. Then the second hidden layer, which consisted of 64 neurons and ReLU activation further refined the inputs. The model's output layer contains a single neuron and a linear activation function, which we needed to predict our market value. One of the key things that we did to visualize and assess the model was plotting its learning behavior. A plot of the learning curve was generated by visualizing the training and validation lost during the training process. MAE was also recorded for each epoch.

TABLE IV
MODEL PARAMETERS FOR SEQUENTIAL DEEP LEARNING

Parameter	Values
input layer	[128, activation 'relu', input dim= # of features(53)]
first hidden layer	[0.2]
second hidden layer	[64, activation='relu']
output	[1, activation='linear']

A stacking ensemble was also constructed. The hope was to combine the strengths of individual models. The meta-model chosen was Linear Regression, with base models including Random Forest, Gradient Boosting, and the neural network. Predictions from the base models were used as input features for the meta-model, blending their outputs to enhance overall performance. Voting regressor was also implemented. It is another ensemble learning technique which combines predictions of multiple models similar to stacking. However, instead of using a meta-model to combine predictions (which was done in stacking), voting averages the predictions of all base models to generate the final prediction.

VII. FEATURE IMPORTANCE ANALYSIS

Feature selection was guided by domain knowledge. However, feature importance is a key factor to understanding our data. With understanding data, we can better predict our targets. Majewski, S (2016) showed that there are many independent variables having an impact on the value of the player. [12] The models can be used to find them. Feature importance via Random Forest, Correlation Heatmap and Recursive feature elimination (RFE) were used to identify the most critical features for predicting market value. Heatmap was already used before to do the bi-variate data analysis, but it did provide important features through basic statistical methods without the use of a model. McHale, I. G., and Holmes, B. (2023) stated how Linear regression can be used to model the log of market value as a function of a collection of player characteristics. [13] So in the future work, we can implement feature selection through a possible best fit line graph visual through linear regression.

VIII. RESULTS

The results of this study highlight the performance of various regression models and ensemble methods applied to soccer player market value prediction. This section provides a detailed account of the performance metrics, observations, and insights derived from each approach.

A. Linear Regression

Linear regression was implemented as a baseline approach. Initially, we utilized the default implementation provided by the Scikit-learn library without conducting feature selection. This approach yielded a promising R^2 score of 0.97, indicating that 97% of the variance in player market values was explained by the model. The MAE was 0.18, and the RMSE was 0.23, demonstrating moderate predictive accuracy. Ridge and Lasso regression were employed to address potential overfitting and improve regularization. Hyperparameter tuning for both methods involved varying the alpha parameter, which controls the regularization strength. The best performance for Ridge regression was observed at an alpha value of 0.004642, yielding an MAE of 0.1697, an RMSE of 0.2269, and an R^2 score of 0.973. These metrics demonstrate a slight improvement over standard linear regression. For Lasso regression, the optimal alpha was determined to be 0.0001. This configuration produced an MAE of 0.1692, an RMSE of 0.2272, and an R^2 score of 0.973. The Lasso results were comparable to those of Ridge regression, affirming the utility of regularization in improving model generalizability.

TABLE V
BEST RIDGE PARAMETERS

Parameter	Values
Alpha	0.004642
MAE	0.169702
RMSE	0.226939
R^2	0.973164

TABLE VI
BEST LASSO PARAMETERS

Parameter	Values
Alpha	0.000100
MAE	0.169714
RMSE	0.227218
R^2	0.973098

Both Ridge and Lasso performed similarly, with close R^2 values. This suggests that regularization technique did not have a large effect on the ability of the models to explain variance in the soccer player's market value. The main difference is in Alpha values. This shows how Ridge tends to work better with slightly higher regularization, while Lasso requires much smaller regularization.

To align with the project guidelines emphasized by Professor Weiss, we explored feature selection to remove irrelevant features. However, before incorporating this guidance, we built a custom implementation of linear regression without

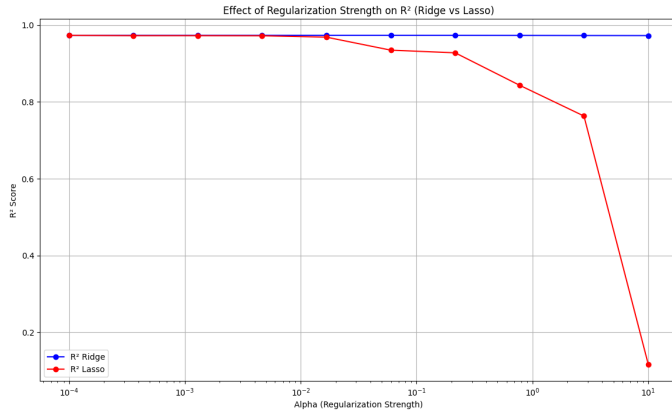


Fig. 3. Effect of Regularization Strength on R²

feature selection. This custom model resulted in a significantly higher MAE of 4.86, indicating that omitting feature selection negatively impacts predictive performance. Upon integrating feature selection, the custom linear regression model's MAE improved dramatically to 0.86. These results underscore the importance of eliminating irrelevant features in regression models, as articulated in the course guidelines.

B. Random Forest

Random Forest demonstrated exceptional performance in predicting soccer player market values, achieving a Mean Absolute Error (MAE) of 0.05, a Root Mean Squared Error (RMSE) of 0.09, and a perfect R² score of 1.00 on the test set. These metrics reflect the model's capacity to capture the complex relationships between features with remarkable accuracy. To ensure the model's generalizability, cross-validation was conducted, yielding a cross-validated RMSE of 0.10. The consistency between the test and cross-validation RMSE values confirms that the model is not significantly overfitting, making it a robust predictor for unseen data. The Random Forest model was further refined using a grid search to explore various hyperparameter combinations as stated in the methodology (Table 1).

And it was found that the optimal parameter configuration was identified as shown in table 7:

TABLE VII
OPTIMAL PARAMETERS FOR RANDOM FOREST

Parameter	Values
Trees	200
max depth	20
min samples split	2
min samples leaf	2
max features	None

The performance:

This fine-tuned configuration resulted in the same MAE (0.05) and RMSE (0.09) as the default setup. The minimal improvement post-tuning suggests that the Random Forest's default settings are inherently well-suited for the given dataset.

TABLE VIII
RANDOM FOREST PERFORMANCE

Metric	Value
MAE	0.05
RMSE	0.09
R ²	1.00

This robustness is one of the model's key advantages, as it performs reliably without extensive hyperparameter adjustments.

The MAE of 0.05 indicates that, on average, the model's predictions deviated only 0.05 units from actual player market values, showcasing exceptional accuracy. The low RMSE of 0.09 further highlights the minimal spread of prediction errors. An R² score of 1.00 suggests that the model explains nearly all the variance in the target variable, making it highly reliable. Overfitting was also tested in the Random Forest Module, and the cross-validated RMSE was 0.10. The negligible difference between test and cross-validation RMSE values (0.09 vs. 0.10) confirms that the model generalizes well to unseen data.

Random Forest outperformed simpler models like Linear Regression and regularized methods such as Ridge and Lasso in all metrics. Compared to Gradient Boosting (MAE: 0.08) and Neural Networks (MAE: 0.25), Random Forest demonstrated superior accuracy and efficiency. The ensemble's ability to achieve near-perfect predictions with default settings further emphasizes its adaptability to diverse datasets.

C. Gradient Boosting

Gradient Boosting emerged as a strong contender in predicting soccer player market values, achieving the results listed in table 9.

TABLE IX
GRADIENT BOOSTING PERFORMANCE

Metric	Value
MAE	0.08
RMSE	0.10
R ²	0.99

These metrics indicate that the model captured a significant proportion of the variance in the target variable, with predictions deviating by an average of 0.08 units from the true market values. While slightly less accurate than Random Forest, Gradient Boosting demonstrated its effectiveness in handling.

Comparing Gradient Boosting with other models, it did quite well. While Gradient Boosting's MAE (0.08) and RMSE (0.10) were slightly higher than Random Forest's (MAE: 0.05, RMSE: 0.09), its performance was competitive. The slight gap may be attributed to the deterministic, sequential nature of Gradient Boosting compared to Random Forest's randomized approach, which may have better leveraged the dataset's diversity. Gradient Boosting outperformed Neural Networks (MAE: 0.25, RMSE: 0.31) significantly, highlighting

its superior efficiency in capturing structured relationships in tabular data. More on neural networks soon.

Although Gradient Boosting inherently balances bias and variance, further performance improvements could have been explored through parameter tuning. The learning rate could have been adjusted to lead to differing step sizes for weight updates. This would balance convergence speed and accuracy. The number of trees could be increased, in turn increasing the number of boosting rounds to capture finer details and unknown patterns. The tree depth can either increase or decrease resulting in differing model complexities. If we lower the number then we could increase the generalization strength.

The MAE of 0.08 suggests that the model made highly accurate predictions, with minimal average deviation from actual values. The RMSE of 0.10 corroborates this accuracy by reflecting a low error spread across predictions. The R^2 score of 0.99 indicates that Gradient Boosting explained 99% of the variance in player market values. This highlights the model’s capacity to capture intricate feature interactions while maintaining generalization. Gradient Boosting consistently delivered robust results across different training-validation splits, reinforcing its reliability.

D. Neural Networks

There were two main implementations for neural networks. The first neural network was implemented as a non-linear regression approach to capture complex patterns in the soccer player market value dataset. The initial configuration, using a single hidden layer with 100 neurons and ReLU activation, produced the results seen in table 10.

TABLE X
INITIAL NEURAL NETWORK PERFORMANCE

Metric	Value
MAE	0.25
RMSE	0.31
R^2	0.95

While these metrics suggest reasonable predictive performance, the neural network underperformed compared to ensemble methods such as Random Forest and Gradient Boosting. The relatively higher error rates indicate that the model struggled to match the accuracy of tree-based methods, possibly due to its sensitivity to hyperparameter settings and the dataset’s characteristics.

Next hyper-parameter tuning was implemented. To improve performance, an extensive grid search was conducted. (See figure 8)

The goal of this tuning process was to identify the best network architecture and training configuration for the dataset. Despite these efforts, the fine-tuned model exhibited significantly worse performance, with the metrics in figure 13.

These results suggest that the tuning process encountered convergence issues or overfitting, causing the model’s performance to degrade instead of improving. The baseline neural network demonstrated some ability to model the relationship

TABLE XI
HYPER PARAMETERIZED NEURAL NETWORK PERFORMANCE

Metric	Value
MAE	2.15
RMSE	3.06
R^2	-3.72

between input features and target values, as indicated by the MAE of 0.25 and an R^2 score of 0.95. But the hyper parameterized models decided to hit very hard. The sharp increase in MAE and RMSE and the negative R^2 score post-tuning suggest that the network overfit the training data or failed to converge effectively.

The likely contributions to the model’s limitation can range from a multitude of things. An overly complex architecture (e.g., deeper or wider layers than needed), poor weight initialization leading to suboptimal convergence, learning rate too high or too low, preventing effective parameter updates, and regularization strength that either penalized weights excessively or insufficiently controlled complexity. Another key issue could be that the dataset was not big enough for the model to learn any intricate patterns without overfitting.

While neural networks have the potential to outperform other models, especially in high-dimensional and unstructured data, their reliance on extensive training and optimization proved to be a disadvantage in this context. Random Forest and Gradient Boosting not only achieved lower MAE and RMSE values but also required less computational effort and tuning.

E. Stacking and Voting Regressor

After already deploying several ensemble methods in this study, the combined the strengths of multiple base models—Random Forest, Gradient Boosting, and Neural Networks—can be used to enhance predictive performance. Two ensemble approaches, Stacking Model and Voting Regressor, were explored.

TABLE XII
STACKING AND VOTING REGRESSOR PERFORMANCE

Model	MAE
Stacking Model	0.05
Voting Regressor	0.09

The stacking model combined Random Forest, Gradient Boosting, and Neural Network as base learners, with a meta-model trained on the outputs of these learners. By leveraging the complementary strengths of the base models, the stacking approach achieved a MAE of 0.05, matching Random Forest’s performance. The reason behind this performance was likely our diverse base models at first glance. But remember how we already saw each of the models’ base performances. Neural networks and Gradient boosting were lacking behind. Due to the stacking model emphasizing the strengths of each model while mitigating the weaknesses. So the stacking model

would have prioritized Random Forest’s consistency for strong predictions.

It would be right to observe that the stacking model’s performance matches the Random Forest’s MAE, suggesting that the latter dominated the ensemble due to its superior predictive capacity. Also the inclusion of Neural Network likely did not add significant value, as evidenced by its relatively weaker standalone performance as we measured before (MAE: 0.25).

The Voting Regressor, another ensemble method, averaged predictions from Random Forest, Gradient Boosting, and Neural Network to produce final outputs. This approach resulted in an MAE of 0.09, which, while still competitive, was inferior to the stacking model.

Unlike the stacking model, the Voting Regressor assigns equal importance to all base models. This approach dilutes the contributions of stronger models like Random Forest and overweights weaker models like the Neural Network. There was also a lack of a meta-model. Without a meta-model, the Voting Regressor cannot adaptively prioritize the predictions of higher-performing models based on data characteristics. The performance gap between Voting Regressor and Stacking Model underscores the importance of meta-model optimization in combining predictions effectively. While computationally less intensive than stacking, the Voting Regressor’s simplicity limits its potential in leveraging model complementarities.

These two ensemble methods, particularly the stacking model, demonstrated their ability to consolidate the strengths of diverse models into a robust and accurate predictor of soccer player market values. The stacking model’s performance matched that of Random Forest while enhancing generalization through meta-learning. However, the marginal improvement highlights that the choice of base models remains critical, and the added complexity of ensemble methods must be weighed against their benefits in similar prediction tasks.

F. Deep Learning

The initial deep learning model for predicting soccer player market value showed moderate performance. Training over 50 epochs revealed significant improvements in performance metrics during early epochs, demonstrating the network’s capacity to learn effectively from the dataset. Key metrics observed during training are in figure 3.

TABLE XIII
TRAINING AND VALIDATION METRICS PER EPOCH

Epoch	Loss	MAE	Validation Loss	Validation MAE
1	45.63	4.91	1.41	0.95
5	1.07	0.82	0.36	0.48
10	0.44	0.52	0.15	0.30
20	0.08	0.22	0.07	0.20

By Epoch 50, the model converged to a test loss of 0.052 and a test MAE of 0.1574. The decreasing loss and MAE values suggest effective learning, and the gap between training and validation errors remained minimal, implying limited overfitting.

The final test MAE of 0.1574 reflects the model’s ability to predict player market values with an average deviation of approximately 0.16 units from the actual values. The loss metric steadily decreased from 45.63 in the first epoch to 0.052 at the final epoch, indicating successful minimization of the error function. The validation metrics closely followed the training metrics throughout the epochs, confirming that the model generalizes well to unseen data.

In the early learning stages, there was a significant drop in loss and MAE within the first 10 epochs. This underscores the deep learning model’s rapid adaptation to the dataset’s structure. This behavior is characteristic of models trained on well-preprocessed and informative datasets. V. Steve Arrul, P. Subramanian and R. Mafas(2022) However, after epoch 30, the performance gains diminished, indicating that the model had likely learned most of the relevant patterns in the data. Any additional training epochs would yield negligible improvements.

While the deep learning model’s final MAE of 0.1574 is competitive, it lags behind the performance of Random Forest and Gradient Boosting models, which achieved MAEs of 0.05 and 0.08, respectively. This suggests that ensemble methods may offer better adaptability to this specific dataset. However, the deep learning approach remains valuable for its scalability and potential in more complex, high-dimensional datasets.

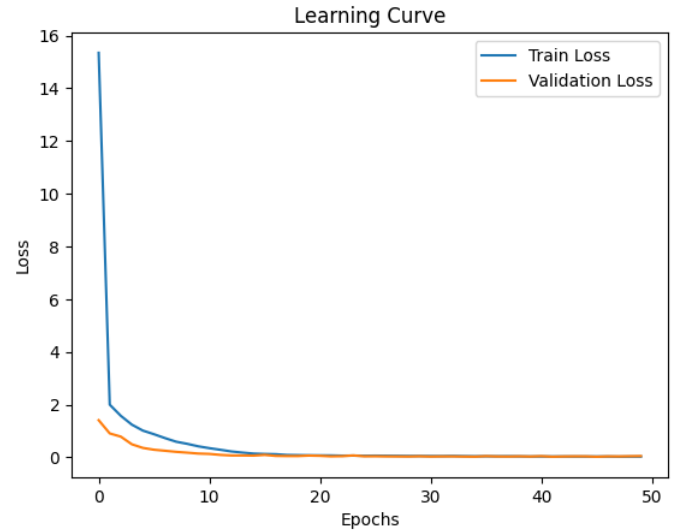


Fig. 4. Learning Curve

G. Feature Importance

Feature importance analysis using Random Forest revealed that overall rating, age to potential ratio, and potential were the top three predictors of market value. Random Forest’s results are seen in table 14. Recursive Feature Elimination (RFE) further refined the feature importance set, identifying the features in table 13. These findings align with domain knowledge, as attributes such as potential and overall rating are intuitively correlated with a player’s market value.

While feature importance was not used directly for feature selection in this study, it was included as a valuable metric to better understand the factors influencing soccer player market values. Random Forest’s built-in feature importance metric and Recursive Feature Elimination (RFE) provided complementary insights into which attributes were most predictive, aligning closely with domain knowledge, supporting the models.

Using Random Forest, the most influential predictors were ranked based on their contribution to reducing model error. The top 15 features are listed in table 14.

TABLE XIV
RANKING OF ATTRIBUTES BY IMPORTANCE VIA RANDOM FOREST

Ranking	Attribute
1	Overall Rating
2	Age-to-Potential Ratio
3	Potential
4	Finishing
5	Performance Index
6	Age
7	Positioning
8	Volleys
9	Sliding Tackle
10	Long Shots
11	Vision
12	Marking
13	Dribbling
14	Standing Tackle
15	Ball Control

Recursive feature elimination was utilized to systematically eliminate less relevant features and refine the set of predictors. The following features were identified as the most impactful:

TABLE XV
RANKING OF ATTRIBUTES BY IMPORTANCE VIA RFE

Ranking	Attribute
1	Age
2	Overall Rating
3	Potential
4	Finishing
5	Volleys
6	Dribbling
7	Positioning
8	Sliding Tackle
9	Age-to-Potential Ratio
10	Performance Index

Random Forest emerged as the most reliable model for predicting soccer player market values. Its combination of accuracy, robustness, and resilience to overfitting made it an invaluable tool for this analysis. The results underscore its suitability for datasets with complex feature interactions and highlight the importance of feature selection and engineering in driving predictive success.

H. Related Work

Predicting the market value of football players is a complex problem with significant implications for club management, player negotiations, and fan engagement. This research builds

upon a growing body of literature exploring data-driven approaches to player valuation. Several key themes emerge from the existing work, informing the methodology and analysis presented in this paper thanks to them.

One prominent theme is the use of machine learning models for prediction. Yiğit, Samak, and Kaya (2020) [14], similar to this research, employed various supervised learning techniques including ridge and lasso regressions, random forests, and extreme gradient boosting. Their work, based on Football Manager data, highlighted the effectiveness of ensemble methods, a finding echoed in our research with the superior performance of Random Forest. However, their focus was on intrinsic player features within a single game environment, while we incorporate a broader range of attributes including real-world performance data. Additionally, our research explores neural networks and a custom deep learning model to assess its potential in enhancing prediction accuracy beyond established techniques.

The choice of data source is another crucial aspect of player value prediction. Several studies, including Mustafa A. Al-Asadi; Sakir Tasdemir (2022) [15] and Behravan and Razavi (2021) [16], utilized data from the FIFA video game series. These studies acknowledged the limitations of using game-generated data, noting the subjectivity of some attributes and the potential influence of real-world transfer fees on in-game ratings. This research acknowledges these limitations and incorporates real-world performance data alongside FIFA attributes to create a more comprehensive and robust prediction model. Furthermore, while Behravan and Razavi (2021) focused on estimating player value in the transfer market using a hybrid PSO-SVR approach, our research targets market value prediction using a wider array of machine learning models. Behravan and Razavi’s use of a hybrid optimization and regression approach, however, offers a potential avenue for future work, particularly in exploring feature selection and parameter tuning for enhanced model performance.

The importance of considering player position is highlighted in both Behravan and Razavi (2021) and Zhang and Kang (2021) [17]. Both papers employed clustering techniques to group players based on their field positions (goalkeeper, defender, midfielder, attacker), recognizing that different skills and attributes contribute differently to the valuation of players in distinct roles. Zhang and Kang’s work employed the XG-Boost method after dimensionality reduction and clustering.

If we had an approach of clustering by position prior to building the prediction models, we could have found much more information. Our player data does cover a multitude of positions.. Our research did extend this concept by specifically analyzing the relative importance of different features, but not within each position group. However, a paper by Gijs Laros [18] did take into account the features that were important to all positions, and features that were important to specific positions. Specifically, we found that potential and overall rating were important for all positions. And for different positions like goalkeepers, shot power was more important for them, but not for someone like defenders. This is true

for all positions. Midfielders market value were correlated also with dribbling. Whilst defenders were correlated with physique. (something we did not have in our data set) This granular analysis provides insights into the unique drivers of market value for each position, contributing to a more nuanced understanding of player valuation.

In our study, we employed Lasso regression as one of the models. It followed the methodology of previous research that utilized various regression methods for player valuation. Yuan He (2020) used multiple models, including Ordinary Least Squares (OLS), K-Nearest Neighbors (KNN), Ridge Regression, and Principal Component Regression (PCR) with different hyperparameter values such as varying k for KNN and λ for Ridge Regression. They applied 10-fold cross-validation for each of their models to ensure robustness in their predictions. [19] Although they used a variety of models, their results did not match the performance levels achieved in our study, with their RMS values reaching 54.7 and 27.27 for overall and respective means, respectively, compared to our significantly lower RMS values of 25.63 and 21.27. These results suggest that while our models outperformed theirs, their approach could still provide valuable insights, and incorporating their models in our analysis could have helped identify whether the performance gap stemmed from the models themselves or limitations in the dataset.

I. Conclusion

This literature was an attempt to make contributions to the field of soccer player market value prediction through machine learning. The comprehensive evaluation of the multi model approach reveals that Random Forest demonstrated superior predictive performance. Random forest achieved exceptional accuracy with a mean absolute error of 0.05 and R^2 score of 1.00. This significantly outperformed the other approaches: neural networks, deep learning, traditional regression methods.

Analysis of feature importance revealed that overall rating, age to potential ratio, and potential are the most crucial predictors of player market value. This aligns with domain expertise. This finding provides valuable insights for stakeholders in the soccer industry, from clubs making transfer decisions to agents creating contracts.

However, certain problems need to be addressed. The dataset represents a snapshot from October 2023. This potentially misses temporal dynamics in player valuation, specifically inflation metrics. We listed several ways of solving this, through using consumer price index or recapturing most recent data. Another thing that could have been implemented was clustering or splitting of the data set. Many authors stated how different roles in soccer, the field positions, have different attributes which correlate higher than others to market value. If we implemented this technique, we would have received even more interesting findings. Our project also only focused on numerical features. We did not take categorical features like team affiliations into account. Many domain experts state that team affiliations and team history does impact market value of a player.

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